

Trajectory Classification with LSTM for Robotic Peg-in-Hole

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UE5 Apprentissage

Master's Machine Learning
Project – 2025–2026





Project Objectives

- ▶ Build a trajectory-level classifier for a 7-DoF peg-in-hole task with **3 classes**.
- ▶ Use LSTM architecture.
- ▶ Deliver two outcomes:
 - ▶ strong performance on the internal test split,
 - ▶ Analysis of **feature influence** on performance.



Problem Framing

- ▶ Input: one full trajectory tensor $\in \mathbb{R}^{150 \times 20}$.
- ▶ Output: one class label $\in \{0, 1, 2\}$ for the complete sequence.
- ▶ Why sequence learning: class information is encoded in **temporal dynamics**, not only static snapshots.
- ▶ Evaluation context: internal split (train/val/test), plus compatibility with an external test set.



Roadmap

1. Data extraction and preparation
2. Model
3. Experimental plan (hyperparameters)
4. Challenges encountered
5. Results and analysis
6. Feature consequences study
7. Conclusion



Dataset and Feature Blocks

- ▶ Files: `class1_trajectories.npy`, `class2_trajectories.npy`, `class3_trajectories.npy`
- ▶ Balanced data: 256 trajectories/class, total 768 trajectories.
- ▶ Shape per sample: 150 timesteps, 20 features.
- ▶ Labels: `class1` $\rightarrow$ 0, `class2` $\rightarrow$ 1, `class3` $\rightarrow$ 2.

Block	Meaning	Size
<code>peg_pose</code>	peg position + quaternion	7
<code>peg_vel</code>	linear + angular velocity	6
<code>hole_pos</code>	hole position + quaternion	7
Total		20



Preprocessing Pipeline

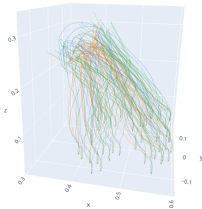
- ▶ Reproducibility: fixed random seed (SEED=42) for split and initialization.
- ▶ Split: **80/10/10** $\Rightarrow$ train 614, val 76, test 78.
- ▶ Normalization: z-score computed on train only, then reused on val/test.
- ▶ Numerical stability: $\sigma = 0$ replaced by small ϵ .

$$\mathbf{x}_{norm} = \frac{\mathbf{x} - \boldsymbol{\mu}_{train}}{\boldsymbol{\sigma}_{train} + \epsilon}$$

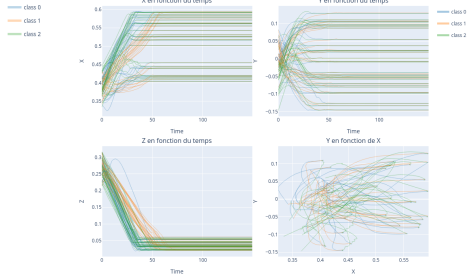


Trajectory Evidence Before Training

Trajectoires (position du peg) par classe



Trajectoires par composante



3D trajectories (x, y, z) by class.

Projections: $x(t), y(t), z(t), y(x)$.

Class-dependent temporal patterns are visible before training, which motivates a temporal classifier.



Why LSTM (Course Connection)

- ▶ Reminder: vanilla RNNs are sensitive to vanishing/exploding gradients through BPTT.
- ▶ LSTM uses gated memory to control information flow and preserve useful long-range signals.

$$i_t, f_t, o_t = \sigma(\cdot), \quad \tilde{c}_t = \tanh(\cdot), \quad c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

- ▶ This architecture matches the project setting: long trajectories (150 steps) with dynamic class signatures.



Final Architecture and Training Setup

- ▶ Input tensor: (batch, $n_{timesteps} = 150$, $n_{features} = 20$).
- ▶ Backbone: LSTM(num_layers=2, hidden_size=64, dropout=0.2).
- ▶ Classifier head: last hidden state $\rightarrow$ Linear(64,3).
- ▶ Loss and optimizer: CrossEntropyLoss + Adam(lr=1e-3).
- ▶ Batch size: 32, training budget up to 40 epochs in main experiments.
- ▶ Stability control: gradient clipping $\|\nabla\|_{max} = 1.0$.



Controlled Experiment Plan

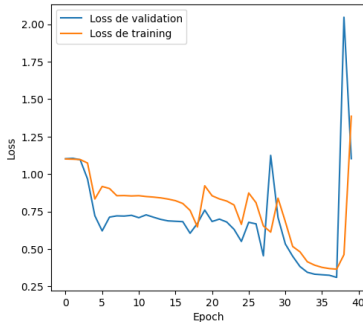
Run	Setting	Purpose
R1	hidden=64, 2 layers, 40 ep, no norm / dropout / clipping	raw baseline
R2	+ z-score + dropout(0.2)	normalization + regularization
R3	R2 + gradient clipping	training stability

- ▶ Fixed controls: seed, split protocol, optimizer family, class definition.
- ▶ Selection criterion: best generalization with stable learning curves.

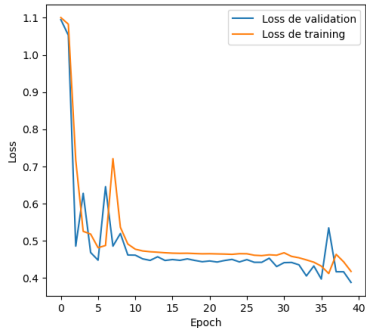


Practical Issues During Development

- ▶ **Reproducibility:** no fixed seed $\Rightarrow$ difficult run-to-run comparison.
- ▶ **Optimization sensitivity:** preprocessing and regularization strongly alter convergence quality.



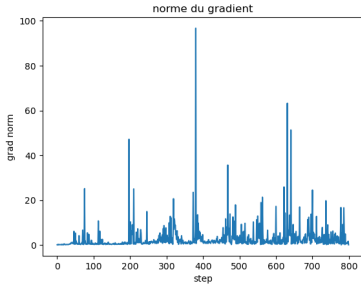
No normalization: noisier and less predictable convergence.



No dropout: weaker regularization and higher instability.



Exploding Gradients and Mitigation



Gradient norm spikes observed during training.

Observed issue

Large gradient spikes destabilize the parameter's updates.

Fix used

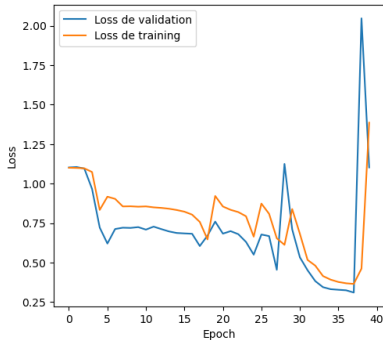
Gradient clipping in the training loop:
`clip_grad_norm(..., 1.0)`.

Effect

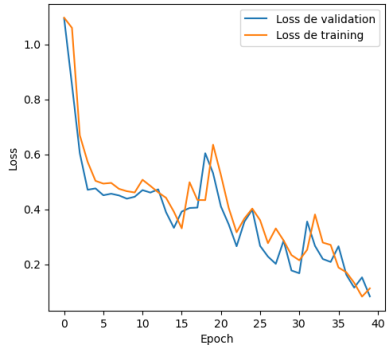
More stable updates and smoother final convergence.



Runs R1 and R2



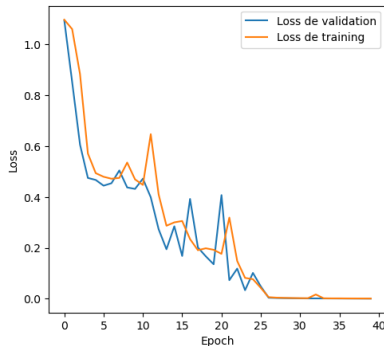
R1 baseline: loss diverges at the end — no normalization, no dropout, no clipping.



R2 (+z-score +dropout): **98.72%** test accuracy (77/78).



Run R3 — Best Run



R3 (+dropout + gradient clipping): **100%** test accuracy (78/78).

- Gradient clipping eliminates late-training spikes and ensures stable convergence.
- This configuration is selected as the final model.



Quantitative Synthesis

Run	Test accuracy	Main effect
R1	diverges	no regularization or normalization
R2	98.72% (77/78)	z-score + dropout improve generalization
R3	100.00% (78/78)	clipping stabilizes training

- ▶ Most impactful additions in this pipeline: **normalization + dropout + clipping**.
- ▶ Internal split performance is excellent; external hidden test remains the final robustness check.



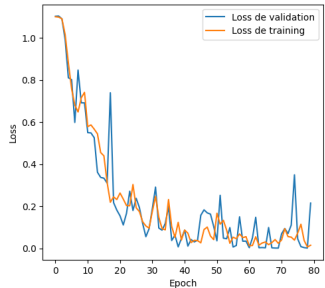
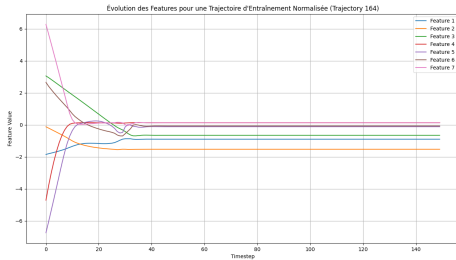
Feature-Ablation Protocol (Feature Notebook)

Subset	Indices	Epoch budgets
Peg pose + orientation	0:7	40, 80, 150
Peg linear+angular velocity	7:13	40, 80, 150
Hole pose + orientation	13:20	40, 150

- ▶ Goal: isolate which observation blocks truly carry discriminative information.
- ▶ Same LSTM family and evaluation protocol; only feature subset and epoch budget vary.



Subset A: Peg Pose/Orientation Only (0:7)



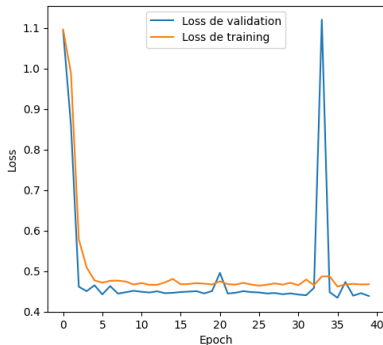
Evolution of the 7 peg pose and orientation features.

Example loss curve (80 epochs).

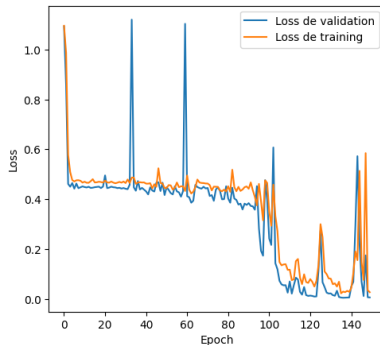
- ▶ Test performance: 98.72% (40 ep), 100% (80 ep), 100% (150 ep).
- ▶ Interpretation: The peg's pose and orientation alone already contain most class-discriminative signal.



Subset B: Peg Velocities Only (7:13)



40 epochs: moderate, unstable plateau.

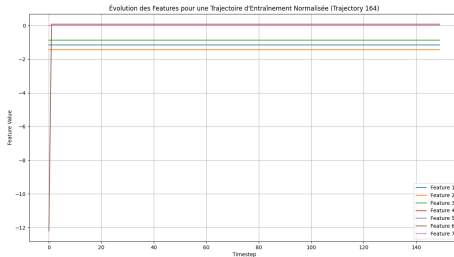


150 epochs: late transition to low loss.

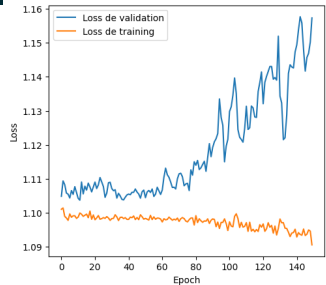
- ▶ Test performance: 64.10% (40 ep), 66.67% (80 ep), 100% (150 ep).
- ▶ Two regimes: long plateau, then sharp improvement around epochs ~ 95 –110.
- ▶ Velocity is informative, but optimization is harder and slower.



Subset C: Hole Pose/Orientation Only (13:20)



Hole-pose features remain constant over time.



150 epochs: no meaningful learning.

- ▶ Test performance drops: 32.05% (40 ep) → 17.95% (150 ep).
- ▶ Validation loss increases with training: more epochs worsen generalization.
- ▶ Interpretation: this subset is non-discriminative for trajectory classification.



Conclusion

- ▶ The final full-feature LSTM pipeline (2 layers, hidden 64, dropout, clipping) reaches top internal performance.
- ▶ Methodical ablation clarifies causality:
 - ▶ The peg's pose and orientation are highly informative,
 - ▶ The peg velocity is informative but optimization-sensitive,
 - ▶ The hole's pose/orientation alone is not discriminative here.
- ▶ Jury-level takeaway: good accuracy comes from both **signal selection** and **stable optimization**, not from architecture choice alone.
- ▶ Final validation step remains the teacher-held hidden test set.



Backup: Why 2 LSTM Layers and Hidden Size 64?

- ▶ The selected configuration is `num_layers=2`, `hidden_size=64`.
- ▶ It gave the best stability/performance tradeoff in our controlled runs.
- ▶ A stronger baseline (hidden size 100) showed weaker behavior in this project context.
- ▶ With 768 trajectories and 150 timesteps, over-capacity increases instability/overfitting risk.
- ▶ Two recurrent layers keep modeling depth while staying computationally light.



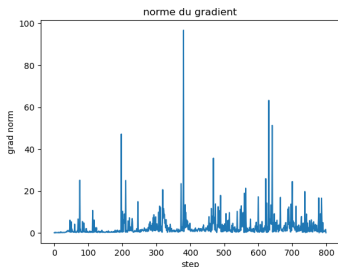
Backup: Model Size and Complexity

Component	Shape/setting	Parameters
LSTM layer 1	input 20, hidden 64	22,016
LSTM layer 2	input 64, hidden 64	33,280
Classifier head	Linear(64,3)	195
Total		55,491

- ▶ The model is compact ($\sim 55k$ params), which is appropriate for this dataset size.
- ▶ Capacity is sufficient to fit complex temporal patterns without requiring very large-scale tuning.



Backup: Why Gradient Clipping at 1.0?



Observed gradient norm spikes before stabilization.

Rule used

`clip_grad_norm(..., 1.0).`

Reasoning

- recurrent training showed large norm spikes
- only large gradients are rescaled

Observed effect

stable late-epoch behavior and no final divergence.



Backup: Feature Interpretation (Physical View)

- ▶ **Peg pose/orientation (0:7)**: directly tracks insertion geometry and trajectory signature.
- ▶ **Peg velocity (7:13)**: informative but noisier; requires longer optimization to fully separate classes.
- ▶ **Hole pose/orientation (13:20)**: near-constant in these trajectories, so weak discriminative power alone.
- ▶ Ablation behavior is therefore physically coherent with the task dynamics.



Backup: Validity Limits and Expected Jury Questions

- ▶ Internal test set size is 78 trajectories: strong result, but still finite-sample uncertainty.
- ▶ Current evidence is mostly split-based; class-wise confusion details can be extended.
- ▶ Single-seed reporting is reproducible but not a full multi-seed robustness study.
- ▶ Final external hidden test remains the decisive out-of-distribution validation.
- ▶ Data leakage risk was mitigated by train-only normalization statistics.



Backup: Why Not GRU or Transformer (Yet)?

- ▶ **Scope fit:** assignment asks for an LSTM trajectory classifier and feature-impact analysis.
- ▶ **Data regime:** 768 sequences is moderate; LSTM is sample-efficient for this scale.
- ▶ **Sequence length:** 150 steps does not force transformer-scale context handling.
- ▶ **Engineering tradeoff:** LSTM gave high accuracy with interpretable/stable training controls.
- ▶ GRU/Transformer remain valid extension baselines for future comparison.